NIRU AI Hackathon 2025: Unified Proposal

Project Title: Sheria Platform

Intelligent AI Platform for End-to-End Government Data Management & Citizen Services

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1. TRACK SELECTION

Track: Generative & Agentic AI
Theme: AI for National Prosperity: Leveraging Innovation for Sustainable Development and Security
Focus: Localized and contextualized Large Language Models (LLMs), generative and agentic AI to enhance education, healthcare, and governance

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2. EXECUTIVE SUMMARY

Sheria Platform is a comprehensive AI-powered ecosystem that transforms how government data is managed, validated, accessed, and utilized across Kenya. By combining intelligent document digitization, validation, rule-driven decision support, and predictive analytics into a unified platform, we address the complete data lifecycle challenge facing Kenyan government institutions.

The platform consists of four integrated modules that work together to democratize access to government data and enable evidence-based decision-making at all levels:

1. **Sheria Digitize** - AI-powered document digitization and metadata extraction
2. **Sheria Verify** - Autonomous document validation and fraud detection
3. **Sheria Ask** - Conversational AI agent for data access and rule-driven insights
4. **Sheria Predict** - Domain-driven predictive analytics for proactive governance

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3. THE COMPREHENSIVE PROBLEM

3.1 The Data Lifecycle Gap

Kenyan government institutions face interconnected challenges across the entire data lifecycle:

Stage 1: Data Creation & Capture (Digitization Bottleneck)

- Millions of physical records remain undigitized or poorly digitized
- Manual data entry is time-consuming (80% of digitization time), error-prone (5-10% error rate), and expensive
- Heterogeneous formats (handwritten, typed, PDFs) in multiple languages (English, Kiswahili)
- Current pace: Kenya will need 15+ years to digitize existing backlogs at current rates

Stage 2: Data Validation & Trust (Fraud & Authentication Crisis)

- Estimated KES billions lost annually to document fraud
- Validation takes days/weeks, requiring physical office visits
- No real-time verification mechanisms for citizens or organizations
- 60%+ of document fraud cases involve sophisticated counterfeits that pass manual inspection

Stage 3: Data Access & Utilization (Siloed Knowledge)

- Government data is locked in incompatible systems across 47 counties and hundreds of departments
- Business rules exist as static code or unwritten tribal knowledge
- Citizens cannot get simple answers: "Why is my application delayed?" "Where is my document?"
- Officers lack decision support tools for consistent service delivery
- Queries that should take seconds require hours of manual cross-referencing

Stage 4: Data Intelligence (Insight Poverty)

- Organizations are "data-rich but insight-poor"
- Predictive analytics requires data scientists most institutions cannot afford
- Domain experts (teachers, nurses, administrators) cannot translate knowledge into predictions
- Reactive rather than proactive governance: schools cannot predict dropout risk, clinics cannot forecast stockouts, counties cannot anticipate service demand

3.2 Impact of the Problem

Economic Impact:

- KES 500M+ annually in manual processing costs across government
- Billions lost to document fraud link
- Months of development time for each predictive analytics use case
- Reduced investor confidence due to document fraud and opaque processes

Social Impact:

- Students drop out before intervention is possible
- Health facilities experience preventable stockouts
- Citizens face "black box" government services with no transparency
- Rural communities cannot access validation services
- Trust erosion in government systems

Operational Impact:

- 80% of government digitization staff time spent on manual data entry
- Officers overwhelmed with validation requests
- Service delivery delays of weeks to months
- Resource allocation based on reaction rather than prediction

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4. THE SHERIA PLATFORM SOLUTION

4.1 Solution Overview

Sheria Platform is an integrated AI ecosystem that addresses each stage of the government data lifecycle through four intelligent, autonomous modules that share a common architecture and data infrastructure.

4.2 Unified Architecture

Core Technology Foundation:

- **Localized LLM Core:** Fine-tuned on Kenyan government documents, administrative text, English & Kiswahili
- **Agentic AI Framework:** Autonomous agents that orchestrate complex multi-step workflows
- **Vector Database:** Semantic search and retrieval-augmented generation (RAG)
- **Rule Engine:** Declarative rule management with natural language translation
- **Secure Data Lake:** Centralized, encrypted storage compliant with Kenya Data Protection Act
- **API-First Design:** Modular integration with existing government systems (eCitizen, IFMIS, Huduma Kenya)

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5. PLATFORM MODULES

MODULE 1: Sheria Digitize

Intelligent Document Processing & Metadata Extraction

Purpose: Accelerate government records digitization using AI to automatically extract, structure, and index metadata from scanned documents.

Key Capabilities:

- **Advanced OCR:** Extract text from scanned images, including handwritten content (Tesseract, Google Cloud Vision)
- **LLM-Powered Extraction:** Understand document context, identify key fields (names, dates, IDs, locations)
- **Named Entity Recognition (NER):** Custom models trained on Kenyan names, locations, government entities
- **Document Classification:** Automatically categorize by type and route to appropriate systems
- **Multi-Language Support:** Process English, Kiswahili, and code-switched documents
- **Quality Assurance:** Confidence scoring, automated validation, human-in-the-loop for low-confidence extractions

Target Documents:

- Birth/death certificates
- Land title deeds
- Tax records, business registrations
- Educational certificates
- Healthcare records
- Court documents

Impact:

- 80-90% reduction in manual data entry time
- Process thousands of documents daily vs. hundreds manually
- 95% accuracy for printed text, >85% for handwritten
- <30 seconds per document processing time

MODULE 2: Sheria Verify

Autonomous Document Validation & Fraud Detection

Purpose: Enable instant validation of government-issued documents through AI agents that orchestrate queries across authoritative data sources.

Key Capabilities:

- **Computer Vision & OCR:** Extract information from submitted documents
- **Agentic Validation Workflow:**
  - Autonomous determination of verification pathway for each document type
  - Multi-source data verification (Land Registry, National Registration Bureau, KRA, Ministry of Education)
  - Cross-referencing across government databases
- **Intelligent Fraud Detection:**
  - ML models trained on verified vs. fraudulent document patterns
  - Signature verification, serial number analysis
  - Anomaly detection across temporal patterns
- **Blockchain Certification:** Immutable validation audit trail
- **Real-Time Processing:** Sub-60-second validation for most document types
- **API Integration:** Easy integration for banks, employers, educational institutions

Use Cases:

- **Citizens:** Instant validation of their own documents
- **Banks/Financial Institutions:** KYC verification, loan processing
- **Employers:** Quick credential verification during hiring
- **Educational Institutions:** Certificate verification for admissions
- **Real Estate:** Land title verification before transactions

Impact:

- Validation time: Days/weeks → <1 minute
- 70% reduction in document fraud (first year target)
- 5M+ validation requests annually
- Save government KES 500M+ in manual processing costs

MODULE 3: Sheria Ask

Conversational Rule-Driven AI Agent for Citizen Services

Purpose: Provide a conversational interface that bridges raw data, domain-specific rules, and people who need answers, making government services transparent and accessible.

Key Capabilities:

- **Intelligent Data Retrieval:** Fetch relevant citizen data across systems using natural language queries
- **Natural Language Rule Translation:**
  - Domain experts define rules in simple, declarative format
  - LLM translates rules into executable code
  - Example: "IF document_type = 'Birth_Certificate' AND applicant_age < 18 THEN require_guardian_signature"
- **Agentic Reasoning:**
  - Retrieve → Evaluate → Predict → Explain workflow
  - Combine business rules with historical data patterns
- **Multilingual Interface:** Chat in English, Kiswahili, or Sheng
- **Transparent Explanations:** Every answer is backed by citable rules and data
- **Multi-User Support:**
  - **Citizens:** "Why is my water connection request pending?"
  - **Officers:** "Show me all land applications in Kajiado at high risk of delay"
  - **Policymakers:** "Which clinic will have highest patient load next month?"

Example Interactions:

Citizen Query:

- Q: "Why is my business license delayed?"
- A: "Your application (Ref #BL-456) is pending because the required fire safety inspection report is missing. Average resolution time: 7 days. Contact the county fire department at [contact]."

Officer Query:

- Q: "List all land title applications missing required signatures"
- A: "Found 23 applications: [list with details]. Rules triggered: RULE_005 (Missing Signature Validation). Recommended action: Send notification to applicants."

Impact:

- Reduce service inquiry response time from days to seconds
- 50% reduction in repeat inquiries through clear, actionable answers
- Enable 24/7 citizen engagement without human officers

- Consistent, rule-based service delivery across all government touchpoints

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MODULE 4: Sheria Predict

Domain-Driven Predictive Analytics for Proactive Governance

Purpose: Democratize predictive analytics by allowing domain experts to encode their knowledge as rules, which AI agents combine with organizational data to generate predictions.

Key Capabilities:

- **Data Ingestion Agent:**
    - Process CSV, Excel, databases with descriptive metadata
    - Automatic schema detection, data quality assessment
    - Temporal pattern recognition
- **Domain Rules Engine:**
    - Natural language rule input: "If a student misses >3 consecutive days and has declining grades, predict high dropout risk"
    - Structured patterns: `IF attendance_rate < 75% AND grade_trend = "declining" THEN risk_level = "high"`
    - Rule conflict resolution and version control
- **Predictive Agent:**
    - Combine historical patterns with domain rules
    - Confidence scoring and explanation generation
    - Continuous learning from new data
    - Powered by Claude 4 for advanced reasoning
- **Conversational Prediction Interface:**
    - "Which students are at highest dropout risk this term?"
    - "Predict vaccine requirements for next month"
    - "Show areas likely to need water delivery in the next dry season"

Sector Applications:

A. Education:

- Student dropout prediction (20-30% reduction target)
- Resource allocation forecasting
- Performance prediction for targeted intervention
- Example Rule: "Students with <75% attendance + declining grades + missed parent meetings = HIGH RISK"

B. Healthcare:

- Disease outbreak prediction (seasonal patterns + weather data)
- Stock management (medicine/vaccine requirements)
- Patient flow forecasting for staffing
- Example Rule: "When cases increase 30% week-over-week + rainy season starts = OUTBREAK ALERT"

C. Governance:

- Service demand prediction (water, sanitation, social services)
- Revenue forecasting for budget planning
- Citizen engagement needs
- Example Rule: "During dry season months + high population density + previous year patterns = INCREASE SERVICE CAPACITY"

Impact:

- 20-30% reduction in student dropout rates
- 15-25% reduction in medicine stockouts
- 30-40% improvement in service delivery efficiency
- Predictive models deployed in days, not months

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6. HOW THE MODULES WORK TOGETHER

6.1 Integrated Data Flow

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Physical Documents → Sheria Digitize → Structured Data
                        ↓
               Metadata Lake
                        ↓
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  ↓                 ↓                 ↓
Sheria Verify      Sheria Ask        Sheria Predict
(Validation)     (Access & Rules)      (Forecasting)
  ↓                 ↓                 ↓
Blockchain Cert    Citizen Answers      Proactive Insights
```

6.2 Real-World Integrated Scenario

Use Case: Student Certificate Verification & Dropout Prevention

1. **Digitization:** School certificates digitized via Sheria Digitize

    - OCR extracts student names, grades, attendance records
    - Metadata stored in central data lake
2. **Validation:** Employer uses Sheria Verify to validate certificate
    - AI agent queries Ministry of Education database
    - Cross-references student ID, institution, completion date
    - Issues blockchain-verified certificate in 30 seconds
3. **Rule-Driven Access:** School administrator uses Sheria Ask
    - Query: "Show students at risk this term"
    - Agent applies rules: attendance < 75% + declining grades
    - Returns list with explanations and recommended actions
4. **Predictive Analytics:** County education officer uses Sheria Predict
    - Historical data + domain rules generate predictions
    - Forecast: "45 students across 5 schools at high dropout risk"
    - Proactive resource allocation for intervention programs

Result: Integrated platform enables full lifecycle management - from creating verified digital records to using that data for intelligent decision-making.

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17. CONCLUSION

Sheria Platform represents a paradigm shift in how Kenyan government institutions manage, validate, access, and leverage their data. By addressing the complete data lifecycle through an integrated AI ecosystem, we transform the "data-rich, insight-poor" reality into "data-driven, insight-rich" governance.

17.1 Why Sheria Platform Will Succeed

1. Comprehensive Solution: Unlike point solutions that address only one problem, Sheria Platform tackles the entire data challenge, creating network effects where each module enhances the value of others.

2. Proven Technology: Built on established AI technologies (Claude, LangChain, Tesseract) with demonstrated success in similar applications worldwide, adapted for Kenyan context.

3. User-Centered Design: Designed with input from real government users, focusing on practical problems and simple interfaces that non-technical users can adopt immediately.

4. Strategic Alignment: Perfectly aligned with Kenya Vision 2030, Digital Economy Blueprint, and Bottom-Up Economic Transformation Agenda, ensuring government support.

5. Scalable Architecture: Modular, API-first design allows gradual adoption—start with one module, expand to all four—reducing implementation risk.

6. Measurable Impact: Clear, quantifiable metrics (time saved, costs reduced, accuracy improved, lives improved) make ROI calculation straightforward.

17.2 Our Ask

We are participating in the NIRU AI Hackathon to:

1. **Validate** the technical feasibility with a working MVP
2. **Demonstrate** the social impact potential to key stakeholders
3. **Build** partnerships with government, academic, and industry partners
4. **Secure** initial funding for pilot program development
5. **Position** Kenya as a leader in AI-driven governance innovation in Africa

17.3 Vision: AI-Powered Governance for Africa

Sheria Platform is more than a hackathon project—it's a blueprint for how AI can transform public service delivery across Africa. By starting in Kenya with a comprehensive, integrated approach, we can create a replicable model that other countries can adapt, positioning Africa as an innovator rather than an adopter of AI technology.

We're not just building software—we're building the foundation for data-driven, transparent, and proactive governance that truly serves citizens.
